



**EAST WEST UNIVERSITY**

**CSE347: Information System Analysis and Design  
(SectionNo.03)**

**Project Report**

**-Smart Transit Navigation System**

**Group no.7**

**Submitted By:**

| <b>Student Name</b> | <b>Student Id</b> |
|---------------------|-------------------|
| MD Sabir Islam Rafi | 2023-3-60-553     |
| MD Sirajul Islam    | 2024-1-60-291     |
| MD Sowad Inam       | 2024-1-60-025     |
| Mohtasim Haque      | 2023-3-60-439     |
| MD Solayman Hossain | 2023-3-60-436     |

**Submitted To:**

**Dr. Hosney Jahan**

**Assistant Professor Department of Computer Science & Engineering  
East West University.**

**Date Of Submission: 20.08.2026**

## Table Of Contents

|                                                 |    |
|-------------------------------------------------|----|
| 1. Background / Introduction.....               | 3  |
| 2. Literature Review .....                      | 3  |
| 2.1 Existing Smart Transportation Systems ..... | 3  |
| 2.2 GPS-Based Bus Tracking .....                | 4  |
| 2.3 Passenger Information Systems .....         | 4  |
| 2.4 Transport Management Systems .....          | 4  |
| 2.5 Emergency Communication .....               | 4  |
| 2.6 Gap Identified.....                         | 4  |
| 3. Problem Statement .....                      | 5  |
| 4. Objectives.....                              | 6  |
| 4.1 Main Objective.....                         | 6  |
| 4.2 Specific Objectives.....                    | 6  |
| 5. Proposed System / Methodology.....           | 7  |
| 5.1 User Authentication .....                   | 7  |
| 5.2 Passenger Services .....                    | 7  |
| 5.3 Driver Services .....                       | 7  |
| 5.4 Bus Monitoring .....                        | 7  |
| 5.5 Crowd Monitoring.....                       | 8  |
| 5.6 Vehicle Maintenance .....                   | 8  |
| 5.7 Emergency Management.....                   | 8  |
| 5.8 Authority Monitoring .....                  | 9  |
| 6. Requirement Analysis.....                    | 9  |
| 6.1 Functional Requirements.....                | 9  |
| 6.2 Non-Functional Requirements .....           | 10 |
| 7. Diagrams .....                               | 12 |
| 7.1 Use Case Diagram.....                       | 12 |
| 7.2 Class Diagram .....                         | 13 |
| 7.3 Sequence Diagram.....                       | 14 |

|                                                              |           |
|--------------------------------------------------------------|-----------|
| <b>8. CPM / PERT.....</b>                                    | <b>15</b> |
| 8.1 Project Activities .....                                 | 15        |
| 8.2 Cpm and pert Diagram .....                               | 16        |
| 8.3 Free Float and Total Float table and Critical path ..... | 17        |
| 8.4 Pert Three time Estimate .....                           | 18        |
| <b>9. User Interface / UI Screenshots .....</b>              | <b>19</b> |
| Fig 9.1- Home Page.....                                      | 19        |
| Fig 9.2- Passenger Registration and Log in .....             | 20        |
| Fig 9.4 - Available Bus and Boarding .....                   | 21        |
| Fig 9.5- Bus Schedule .....                                  | 22        |
| Fig 9.6- Notices .....                                       | 22        |
| Fig 9.7- Passenger Dashboard .....                           | 23        |
| Fig 9.8 - Driver login page .....                            | 23        |
| Fig 9.9 - Driver share bus location .....                    | 24        |
| Fig 9.10 - Admin Dashboard .....                             | 24        |
| Fig 9.11 -User Managemnet By Admin.....                      | 25        |
| Fig 9.12 -Bus Management By Admin.....                       | 25        |
| Fig 9.13 - Driver Management By Admin .....                  | 26        |
| Fig 9.14 -Vehicle Fitness .....                              | 26        |
| Fig 9.15 – Reports .....                                     | 27        |
| Fig16- Activity log .....                                    | 27        |
| <b>10. Limitation .....</b>                                  | <b>28</b> |
| <b>11. Future Improvements.....</b>                          | <b>28</b> |
| <b>13. conclusion.....</b>                                   | <b>29</b> |

# 1. Background / Introduction

Public transportation users often face problems such as not knowing the current location of buses, uncertain arrival times, route confusion, delayed service information and slow communication during emergencies. At the same time, transport authorities may struggle to monitor buses, vehicle fitness, passenger density, maintenance, and emergency situations efficiently.

To address these problems, we proposed the **Smart Transit Navigation System**, a centralized digital platform designed to make public transportation information more accessible, connected and manageable. The main idea behind the system is to bring passengers, drivers and transport authorities together on a common platform.

**Passengers** can search routes, track buses through GPS, view estimated arrival information, check crowd density, receive notifications, and access women-only bus information.

**Drivers** can share their GPS location, update vehicle status, report emergencies, and receive route guidance.

**Transport authorities** can manage vehicles, monitor operations, check vehicle fitness, and view maintenance reports.

Our Navigation system used GPS tracking, a centralized database, role-based access, notifications, and web interfaces to provide timely transportation information and improve communication between passengers and transport authorities.

## 2. Literature Review

### 2.1 Existing Smart Transportation Systems

Modern public transportation systems commonly use GPS-based vehicle tracking to provide real-time vehicle locations. Mobile applications and web platforms allow passengers to access route information and estimated arrival times.

## 2.2 GPS-Based Bus Tracking

GPS technology allows the location of a bus to be captured and transmitted to a central system. The Smart Transit Navigation System uses this concept through the `GPSLocation` component connected to the **Bus** actor.

## 2.3 Passenger Information Systems

Existing transit systems provide information such as routes, schedules, vehicle locations, and service announcements. Our system extends this by allowing passengers to also view crowd density, women-only bus information, and emergency notifications.

## 2.4 Transport Management Systems

Transport management platforms generally provide administrators with vehicle and route monitoring facilities. Our system includes **TransportAuthority** roles for vehicle management, fitness monitoring, maintenance reports, and transport operations.

## 2.5 Emergency Communication

Emergency reporting is an important feature in transportation systems. In Our UML, the **Driver** can report an emergency while the **TransportAuthority** can monitor or receive emergency information.

## 2.6 Gap Identified

Many basic transit applications primarily focus on passenger-facing route and location information. Our proposed system combines:

- Passenger information
- Driver communication
- GPS tracking
- Crowd monitoring
- Vehicle fitness
- Maintenance reporting
- Emergency alerts

- Transport authority monitoring

within one centralized system.

### 3. Problem Statement

The existing public transportation process may rely heavily on manual communication and scattered information. Passengers may not know the current location of a bus or its estimated arrival time. They may also have difficulty obtaining updated route or service information.

Drivers may not have an organized mechanism to communicate vehicle conditions or emergency situations to the authority. Transport authorities may also face difficulties monitoring multiple buses, vehicle fitness, maintenance records, passenger density, and emergencies from a centralized platform.

Therefore, there is a need for an integrated system that can:

- Provide real-time bus location information.
- Help passengers find suitable routes.
- Provide estimated arrival information.
- Display passenger/crowd density.
- Provide women-only bus information.
- Allow drivers to report emergencies.
- Monitor vehicle fitness and maintenance.
- Send service and emergency notifications.
- Centralize transport information for authorities.
- Improve communication between passengers, drivers, and authorities.

## 4. Objectives

### 4.1 Main Objective

To develop a **Smart Transit Navigation System** that improves public transportation information, monitoring, communication, and emergency response through a centralized digital platform.

### 4.2 Specific Objectives

1. To provide passengers with updated bus route information.
2. To provide live bus location through GPS.
3. To provide estimated arrival time information.
4. To allow passengers to view crowd density.
5. To provide information about women-only buses.
6. To allow drivers to update bus location and status.
7. To allow drivers to report emergency situations.
8. To allow bus owners to manage vehicles.
9. To monitor vehicle fitness and maintenance information.
10. To allow authorities to monitor transport operations.
11. To provide emergency and service notifications.
12. To maintain centralized transport records.
13. To improve communication between different transport stakeholders.

## 5. Proposed System / Methodology

The Smart Transit Navigation system follows a centralized client-server architecture.

### 5.1 User Authentication

Users register and log into the system. The system identifies their role, such as:

- Passenger
- Driver
- Transport Authority

### 5.2 Passenger Services

The Passenger can:

- Search routes
- Track buses
- View crowd density
- Receive notifications
- Access women-only bus information

### 5.3 Driver Services

The Driver can:

- Share GPS location
- Update vehicle status
- Report emergencies
- Receive route guidance

### 5.4 Bus Monitoring

The Bus stores information such as:

- Bus ID

- Bus number
- Capacity
- Status
- Current location
- Registration number
- Fitness status

## 5.5 Crowd Monitoring

The CrowdMonitor maintains information about:

- Crowd level
- Occupancy

## 5.6 Vehicle Maintenance

The BusOwner can manage vehicles and view maintenance information.

The MaintenanceReport component stores:

- Report ID
- Findings
- Service date

## 5.7 Emergency Management

Drivers can report emergencies. Emergency information includes:

- Emergency ID
- Emergency type
- Location
- Status

## 5.8 Authority Monitoring

The TransportAuthority can:

- Manage users
- Add/edit/delete users
- Manage transport operations
- View transport reports

# 6. Requirement Analysis

## 6.1 Functional Requirements

### 1. Passenger

- Register/login.
- Search routes.
- Track buses.
- View crowd density.
- View women-only bus information.
- Receive notifications.

### 2. Driver

- Login.
- Share GPS location.
- Update vehicle status.
- Report emergencies.
- Receive route guidance.

### **3. Transport Authority**

- Manage users.
- Monitor transport operations.
- Manage system information.
- View transport reports.
- Monitor vehicle fitness and maintenance.

### **4. System**

- Store user information.
- Store bus information.
- Store route information.
- Store GPS information.
- Maintain emergency records.
- Maintain maintenance records.
- Generate notifications.
- Monitor passenger density.

## **6.2 Non-Functional Requirements**

### **1. Performance**

The system provides bus location and service information with minimal delay.

### **2. Security**

The system uses secure authentication and role-based access.

### **3. Reliability**

The system maintains reliable records of buses, routes, emergencies, and maintenance.

### **4. Availability**

The system is available to passengers and authorities whenever transport information is required.

### **5. Usability**

The interface is simple enough for passengers, drivers, and administrators to use.

### **6. Maintainability**

The system allows future updates to routes, buses, users, and system functionality.

### **7. Scalability**

The system is capable of supporting additional buses, routes, users, and transport authorities.

## 7. Diagrams

### 7.1 Use Case Diagram

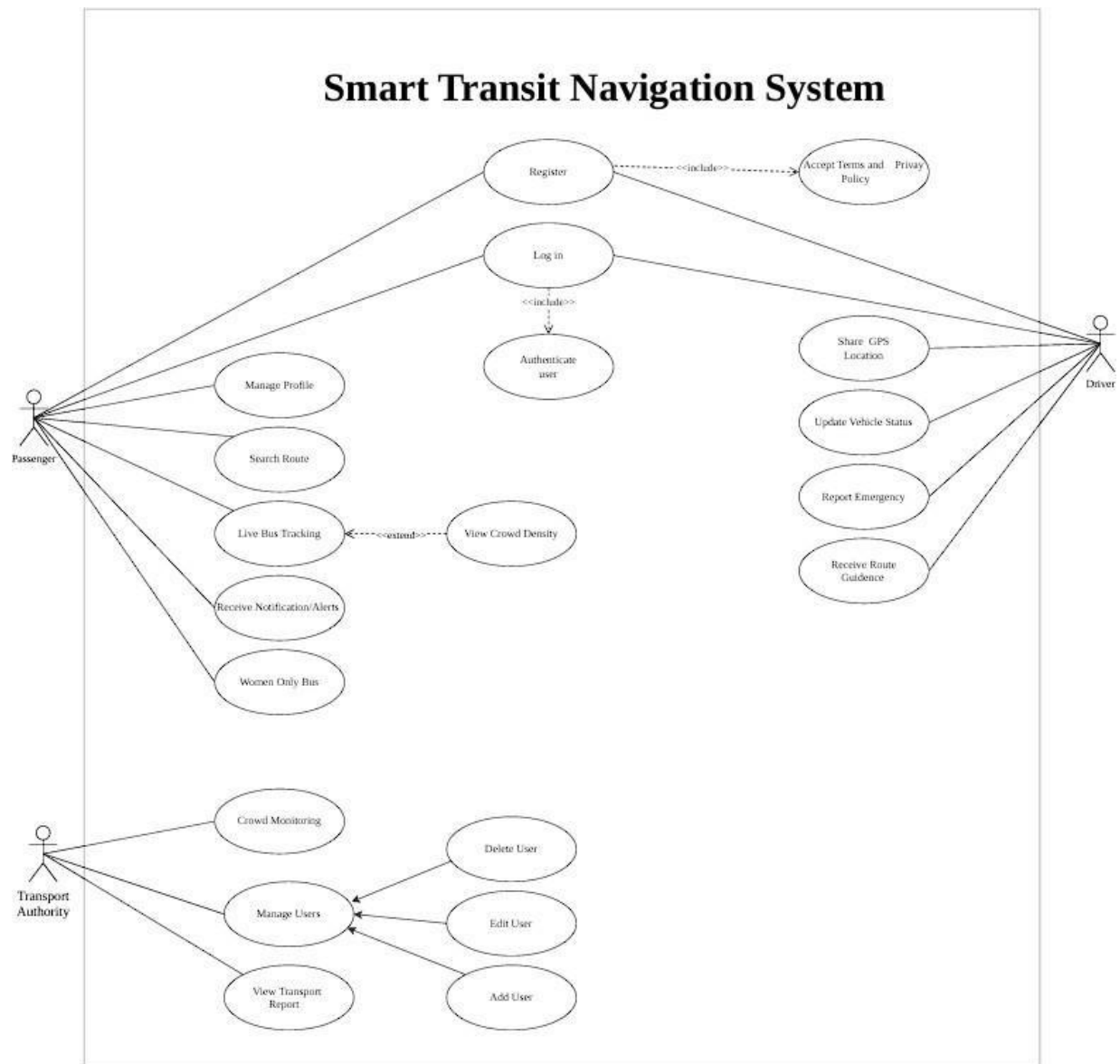


Figure 01: Use Case Diagram of Smart Transit Navigation system

## 7.2 Class Diagram

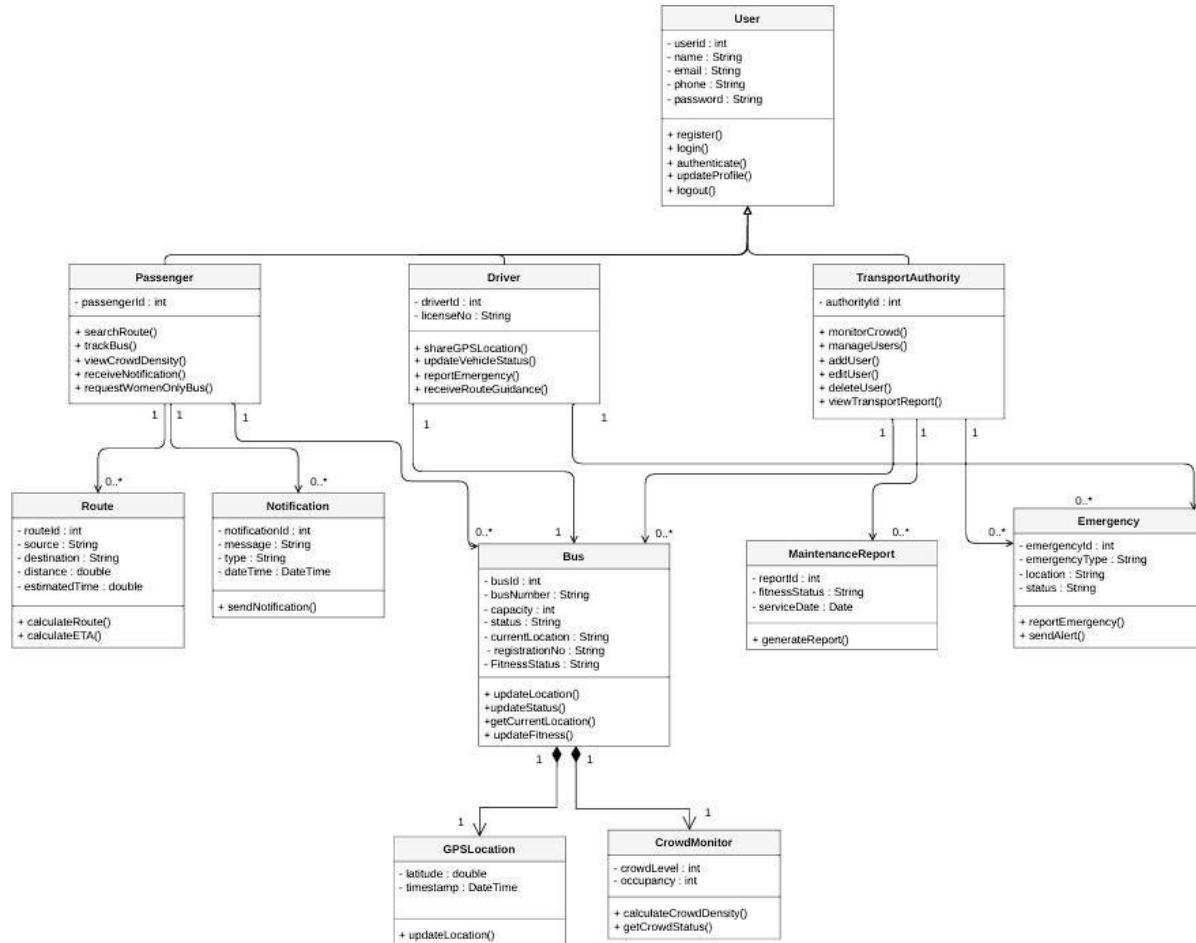


Figure 02: Class Diagram of Smart Transit Navigation system

## 7.3 Sequence Diagram

For Passenger live bus tracking:

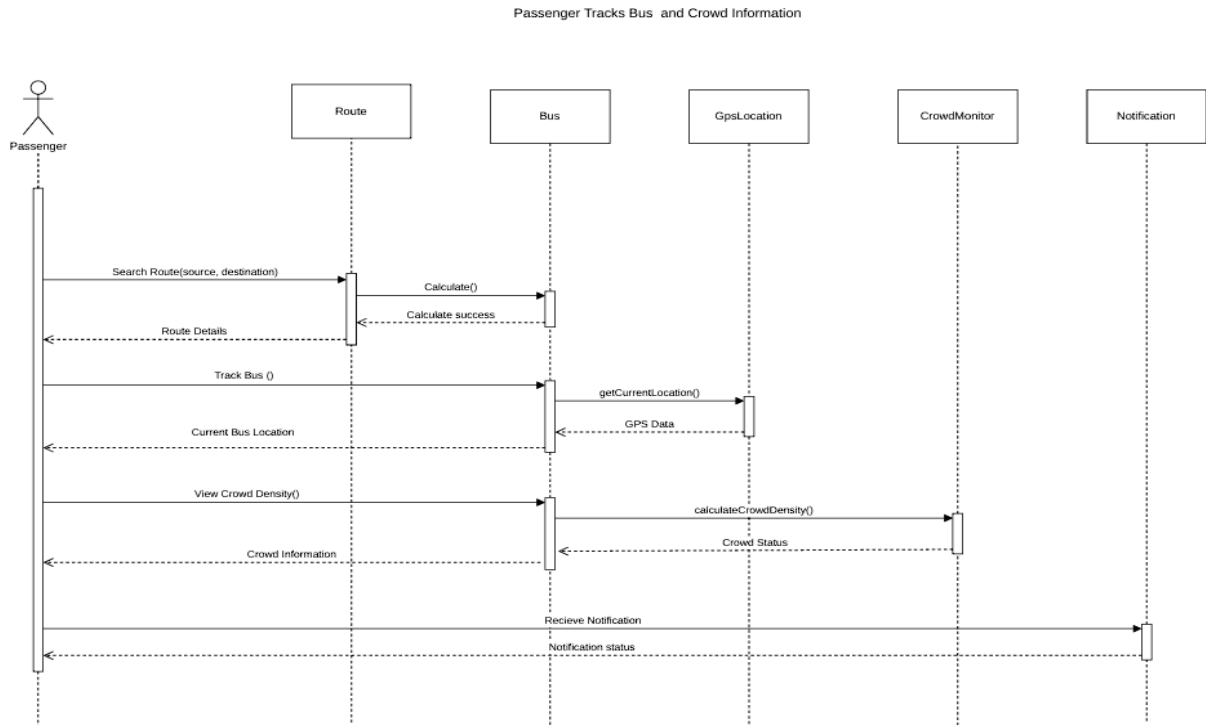


Figure 03: Sequence Diagram of passenger on Smart Transit Navigation system

## 8. CPM / PERT

### 8.1 Project Activities

| Activity                                               | Task ID | Dependency | Days |
|--------------------------------------------------------|---------|------------|------|
| 1. Data collection                                     | A       | None       | 6    |
| 2. Feasibility analysis                                | B       | A          | 3    |
| 3. Requirement analysis                                | C       | B          | 5    |
| 4. SRS Preparation                                     | D       | C          | 4    |
| 5. System design and architecture planning             | E       | D          | 8    |
| 6. Database design for users, buses, routes and alerts | F       | E          | 6    |
| 7. User interface design for app and dashboard         | G       | E          | 5    |
| 8. User authentication and profile module              | H       | F, G       | 7    |
| 9. Route and schedule management module                | I       | F, G       | 8    |
| 10. Live bus tracking using GPS                        | J       | H, I       | 7    |
| 11. Admin dashboard development                        | K       | H, I       | 8    |
| 12. Notification and emergency alert system            | L       | J, K       | 3    |
| 13. Women-only bus information module                  | M       | I          | 3    |
| 14. Vehicle fitness monitoring module                  | N       | K          | 5    |
| 15. Final testing, bug fixing, training and deployment | O       | L, M, N    | 12   |

## 8.2 Cpm and pert Diagram

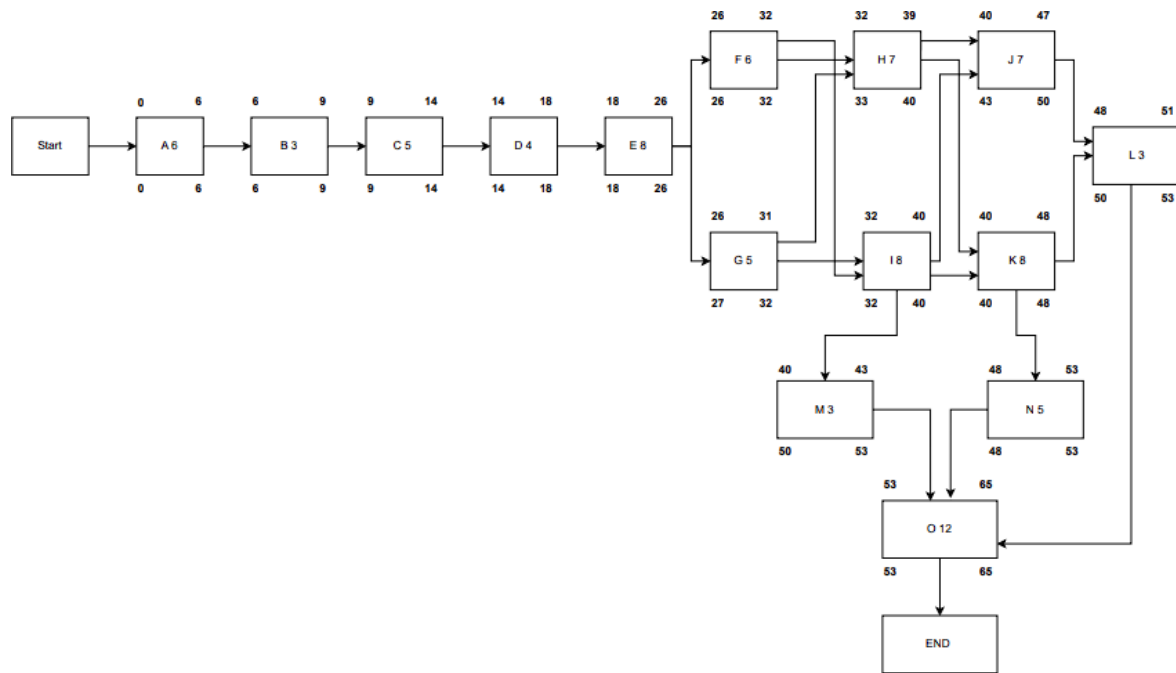


Figure 04: CPM Diagram of Smart Transit Navigation system

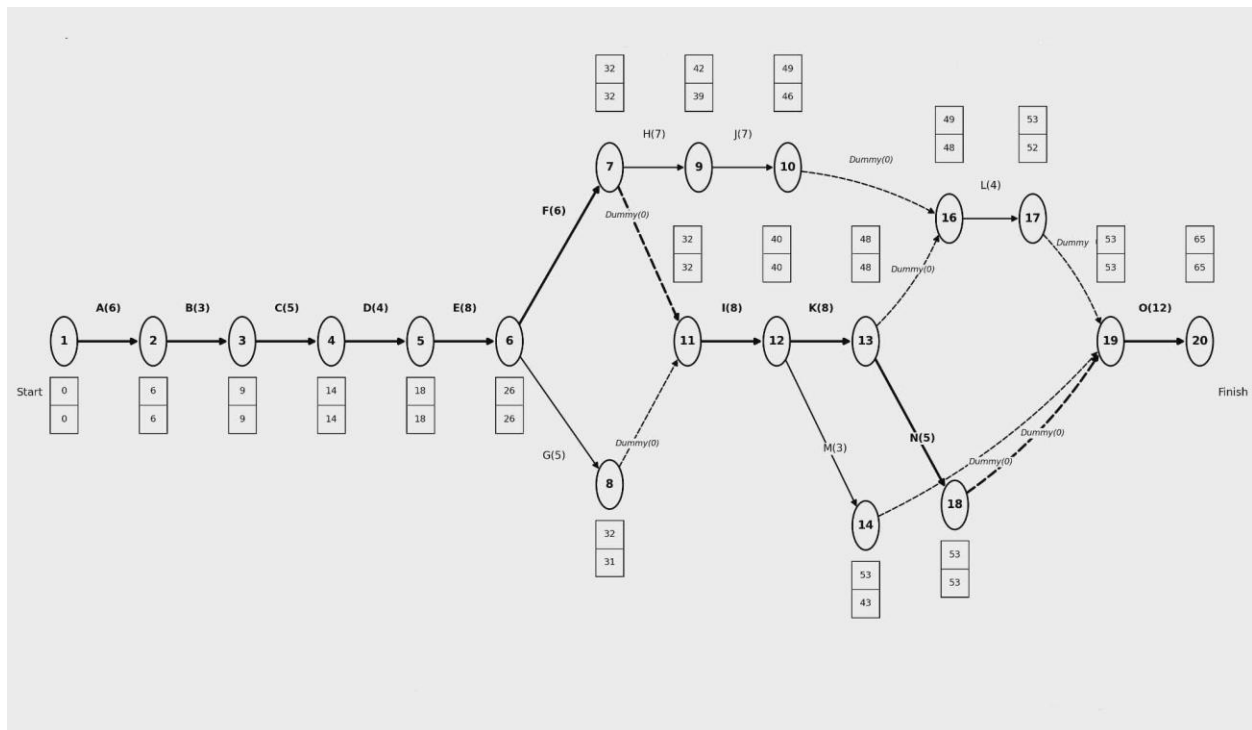


Figure 05: PERT Diagram of Smart Transit Navigation system

### 8.3 Free Float and Total Float table and Critical path

| Activity | Total Float Calculation<br>Total Float = LS – ES = LF – EF | TF | Free Float Calculation : Free float<br>= min ES (next task) – EF (current task) | FF |
|----------|------------------------------------------------------------|----|---------------------------------------------------------------------------------|----|
| <b>A</b> | $0 - 0 = 0$                                                | 0  | $6 - 6 = 0$                                                                     | 0  |
| <b>B</b> | $6 - 6 = 0$                                                | 0  | $9 - 9 = 0$                                                                     | 0  |
| <b>C</b> | $9 - 9 = 0$                                                | 0  | $14 - 14 = 0$                                                                   | 0  |
| <b>D</b> | $14 - 14 = 0$                                              | 0  | $18 - 18 = 0$                                                                   | 0  |
| <b>E</b> | $18 - 18 = 0$                                              | 0  | $26 - 26 = 0$                                                                   | 0  |
| <b>F</b> | $26 - 26 = 0$                                              | 0  | $\min(32,32) - 26 = 6$                                                          | 6  |
| <b>G</b> | $27 - 26 = 1$                                              | 1  | $\min(32,32) - 31 = 1$                                                          | 1  |
| <b>H</b> | $33 - 32 = 1$                                              | 1  | $\min(40,40) - 39 = 1$                                                          | 1  |
| <b>I</b> | $32 - 32 = 0$                                              | 0  | $\min(40,40) - 40 = 0$                                                          | 0  |
| <b>J</b> | $43 - 40 = 3$                                              | 3  | $48 - 47 = 1$                                                                   | 1  |
| <b>K</b> | $40 - 40 = 0$                                              | 0  | $\min(50,48) - 48 = 0$                                                          | 0  |
| <b>L</b> | $50 - 48 = 2$                                              | 2  | $53 - 51 = 2$                                                                   | 2  |
| <b>M</b> | $50 - 40 = 10$                                             | 10 | $53 - 43 = 10$                                                                  | 10 |
| <b>N</b> | $48 - 48 = 0$                                              | 0  | $53 - 53 = 0$                                                                   | 0  |
| <b>O</b> | $53 - 53 = 0$                                              | 0  | $65 - 65 = 0$                                                                   | 0  |

**Critical path:** A→B→C→D→E→F→I→K→N→O = **65 days**

## 8.4 Pert Three time Estimate

| Activity | Dependency | Optimistic (to) | Most Likely (tm) | Pessimistic (tp) | te | $\sigma^2$ |
|----------|------------|-----------------|------------------|------------------|----|------------|
| <b>A</b> | None       | 3               | 6                | 9                | 6  | 1          |
| <b>B</b> | A          | 1               | 3                | 5                | 3  | 0.44       |
| <b>C</b> | B          | 2               | 5                | 8                | 5  | 1          |
| <b>D</b> | C          | 2               | 4                | 10               | 4  | 1.78       |
| <b>E</b> | D          | 4               | 8                | 12               | 8  | 1.78       |
| <b>F</b> | E          | 3               | 6                | 9                | 6  | 1          |
| <b>G</b> | E          | 2               | 5                | 10               | 5  | 1.78       |
| <b>H</b> | F, G       | 4               | 7                | 10               | 7  | 1          |
| <b>I</b> | F, G       | 5               | 8                | 11               | 8  | 1          |
| <b>J</b> | H, I       | 4               | 7                | 10               | 7  | 1          |
| <b>K</b> | H, I       | 5               | 8                | 11               | 8  | 1          |
| <b>L</b> | J, K       | 1               | 3                | 5                | 3  | 0.44       |
| <b>M</b> | I          | 2               | 3                | 8                | 3  | 1          |
| <b>N</b> | K          | 2               | 5                | 10               | 5  | 1.78       |
| <b>O</b> | L, M, N    | 6               | 12               | 18               | 12 | 4          |

The supplied PERT material uses  $te = (to + 4tm + tp) / 6$  and  $\sigma^2 = ((tp - to) / 6)^2$ .

Expected Project Length =  $6+3+5+4+8+6+8+8+5+12 = 65$  days

Project Variance =  $1+0.44+1+1.78+1.78+1+1+1+1.78+4 = 14.78$

Standard Deviation (SD) =  $\sqrt{14.78} = 3.84$

## 9. User Interface / UI Screenshots

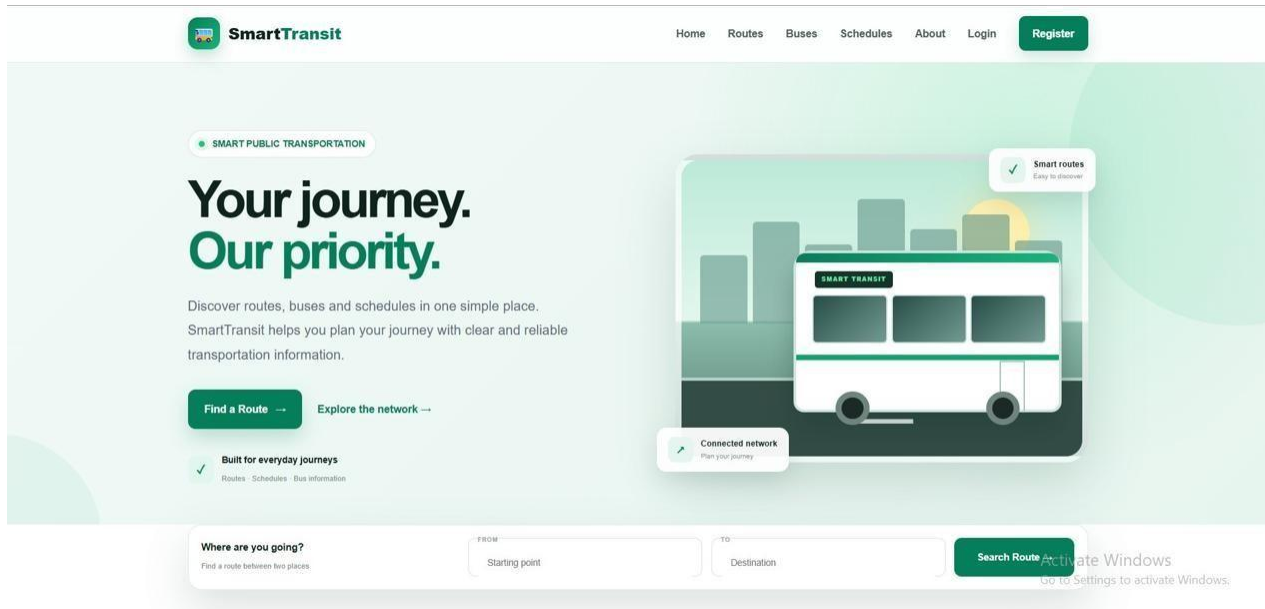
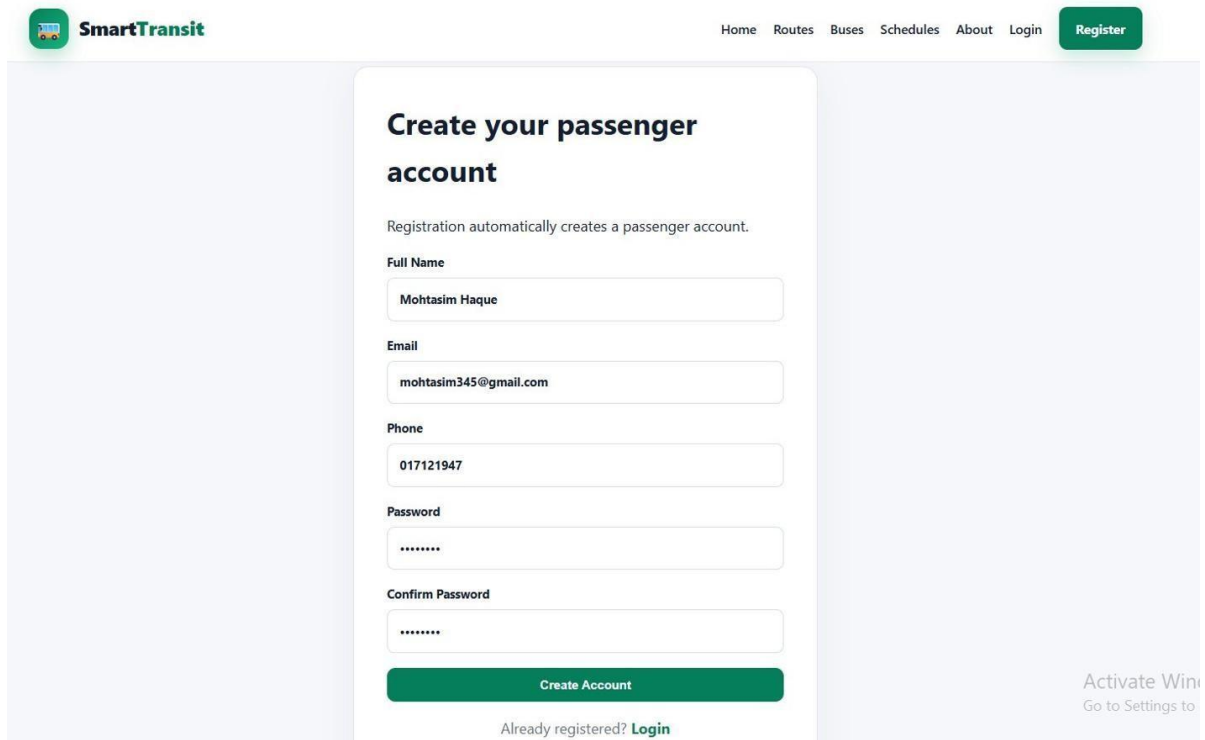


Fig 9.1- Home Page



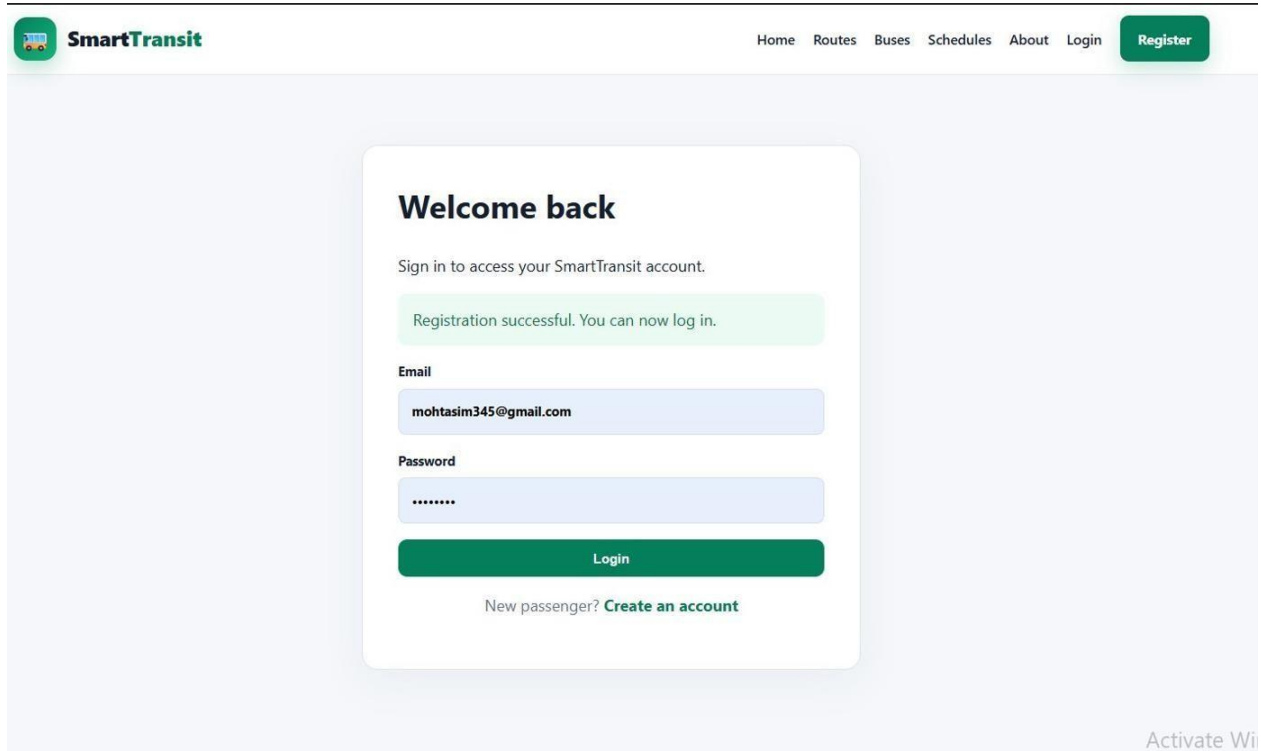


Fig 9.2- Passenger Registration and Log in

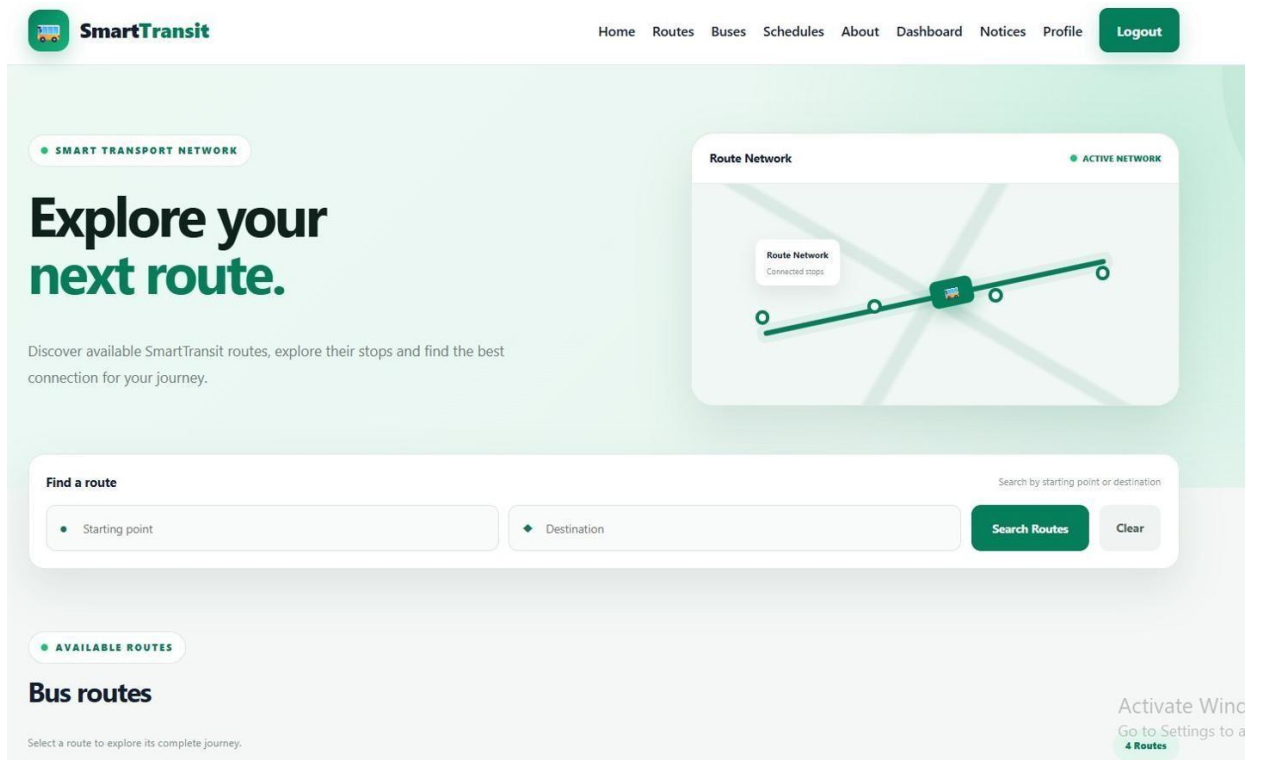


Fig 9.3 - Search route and Available route

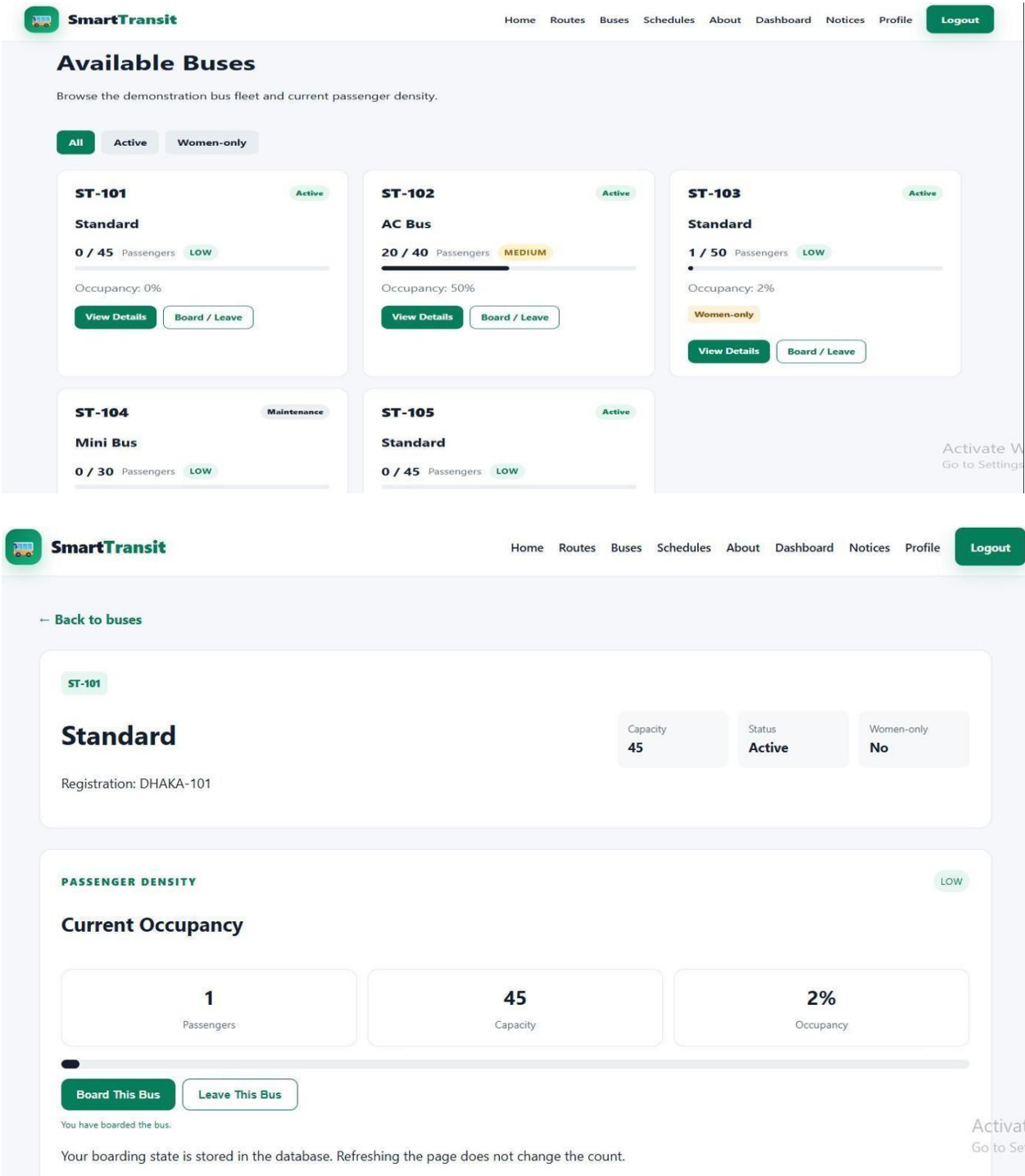


Fig 9.4 - Available Bus and Boarding


SmartTransit
Home Routes Buses Schedules About Dashboard Notices Profile Logout

## TIMETABLE

### Bus Schedules

Check planned route and bus operating times.

All routes
All buses
Any day
Filter

| ROUTE                          | BUS    | DEPARTURE | ARRIVAL  | OPERATING DAYS                                    | STATUS |
|--------------------------------|--------|-----------|----------|---------------------------------------------------|--------|
| R-04 · Dhanmondi to Airport    | ST-105 | 06:30:00  | 08:30:00 | Saturday,Sunday,Monday,Tuesday,Wednesday,Thursday | Active |
| R-01 · Mirpur to Motijheel     | ST-101 | 07:00:00  | 09:00:00 | Saturday,Sunday,Monday,Tuesday,Wednesday,Thursday | Active |
| R-03 · Mohammadpur to Gulistan | ST-103 | 07:30:00  | 09:00:00 | Saturday,Sunday,Monday,Tuesday,Wednesday,Thursday | Active |
| R-02 · Uttara to Farmgate      | ST-102 | 08:00:00  | 09:30:00 | Saturday,Sunday,Monday,Tuesday,Wednesday,Thursday | Active |

Fig 9.5- Bus Schedule


SmartTransit
Home Routes Buses Schedules About Dashboard Notices Profile Logout

## Notices

Important transit information for passengers.

Warning medium priority
**Route information is demonstrational**

The routes and stops currently shown are sample data and are not official transport authority information.

2026-08-19 00:15:55

Warning low priority
**Road Blockage**

Everyone please be seated.

2026-08-19 00:26:37

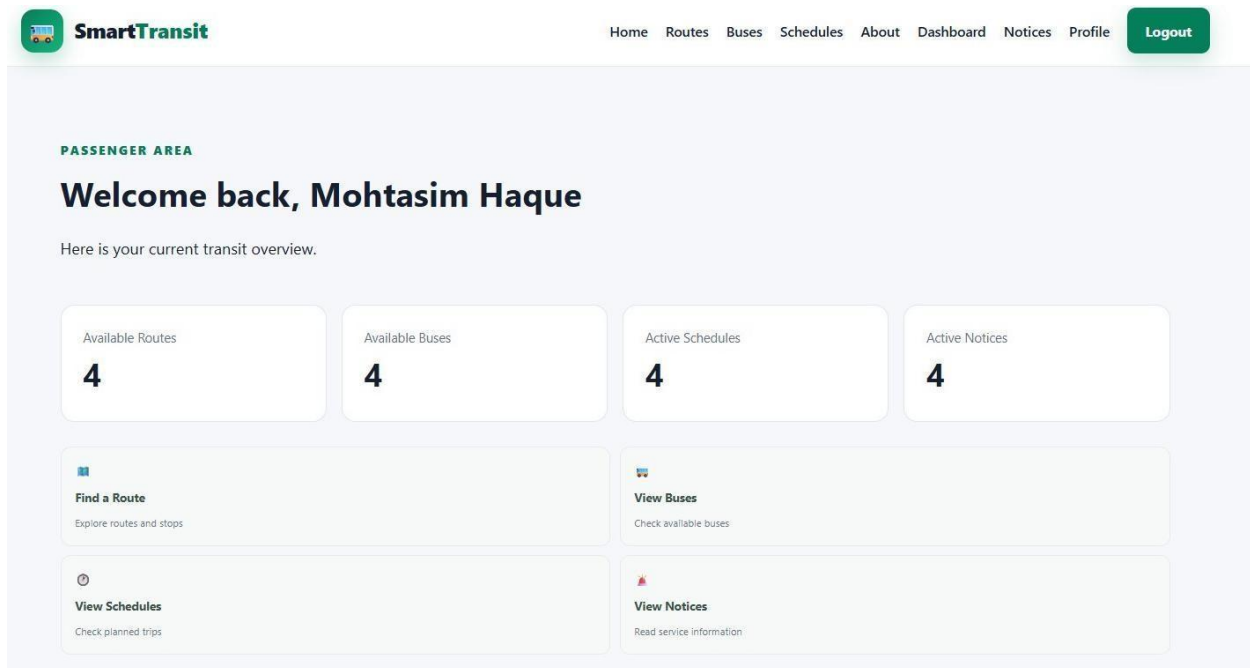
Information low priority
**Welcome to SmartTransit**

This is a demonstration service notice for Part 1 of the project.

2026-08-19 00:15:55

Activate V  
Full Edition

Fig 9.6- Notices



The image shows the 'PASSENGER AREA' dashboard for a user named Mohtasim Haque. The top navigation bar includes links for Home, Routes, Buses, Schedules, About, Dashboard, Notices, Profile, and a Logout button. The dashboard features a welcome message and a summary of transit overview with four key metrics: Available Routes (4), Available Buses (4), Active Schedules (4), and Active Notices (4). Below these are four interactive cards: 'Find a Route' (Explore routes and stops), 'View Buses' (Check available buses), 'View Schedules' (Check planned trips), and 'View Notices' (Read service information).

**PASSENGER AREA**

## Welcome back, Mohtasim Haque

Here is your current transit overview.

| Metric           | Value |
|------------------|-------|
| Available Routes | 4     |
| Available Buses  | 4     |
| Active Schedules | 4     |
| Active Notices   | 4     |

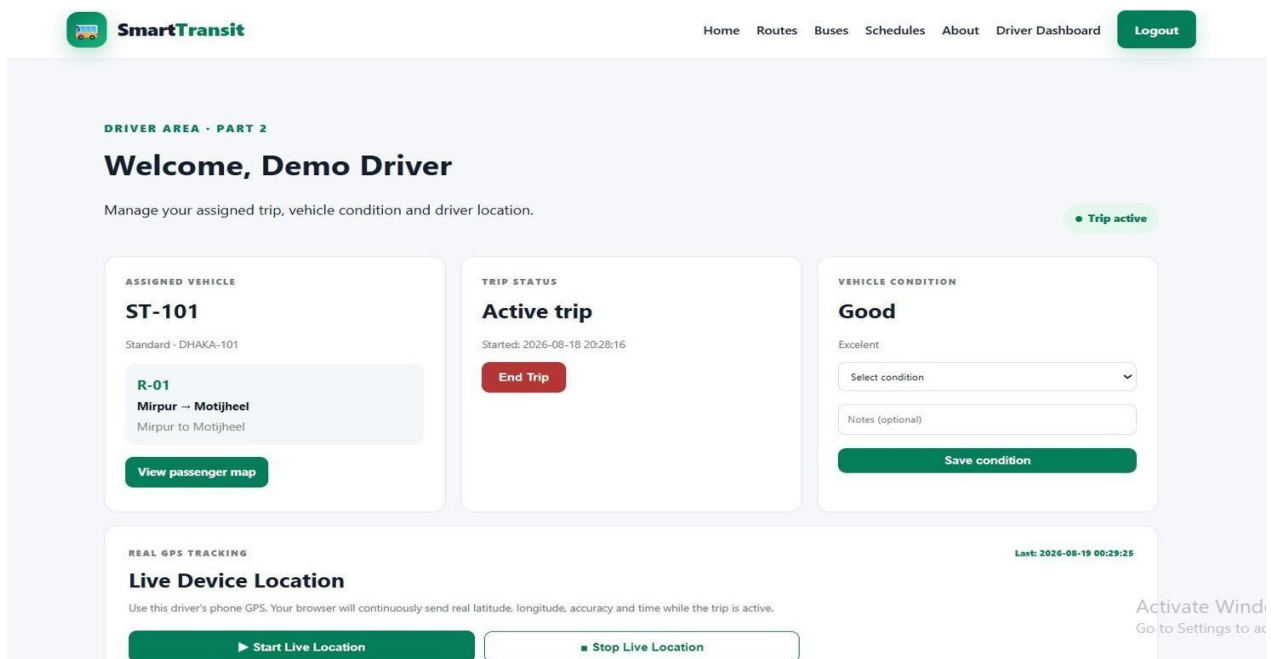
**Find a Route**  
Explore routes and stops

**View Buses**  
Check available buses

**View Schedules**  
Check planned trips

**View Notices**  
Read service information

Fig 9.7- Passenger Dashboard



The image shows the 'DRIVER AREA - PART 2' dashboard for a user named Demo Driver. The top navigation bar includes links for Home, Routes, Buses, Schedules, About, Driver Dashboard, and a Logout button. The dashboard features a welcome message and a summary of driver information. It includes three main sections: 'ASSIGNED VEHICLE' (ST-101, Standard - DHAKA-101, R-01, Mirpur to Motijheel), 'TRIP STATUS' (Active trip, Started: 2026-08-18 20:28:16, End Trip button), and 'VEHICLE CONDITION' (Good, Excellent, Select condition dropdown, Notes (optional), Save condition button). At the bottom, there is a 'REAL GPS TRACKING' section with 'Live Device Location' and buttons for 'Start Live Location' and 'Stop Live Location'. A 'Trip active' status indicator is also present.

**DRIVER AREA - PART 2**

## Welcome, Demo Driver

Manage your assigned trip, vehicle condition and driver location.

**ASSIGNED VEHICLE**

**ST-101**  
Standard - DHAKA-101

**R-01**  
Mirpur to Motijheel  
Mirpur to Motijheel

**View passenger map**

**TRIP STATUS**

**Active trip**  
Started: 2026-08-18 20:28:16

**End Trip**

**VEHICLE CONDITION**

**Good**  
Excellent

Select condition

Notes (optional)

**Save condition**

**REAL GPS TRACKING**

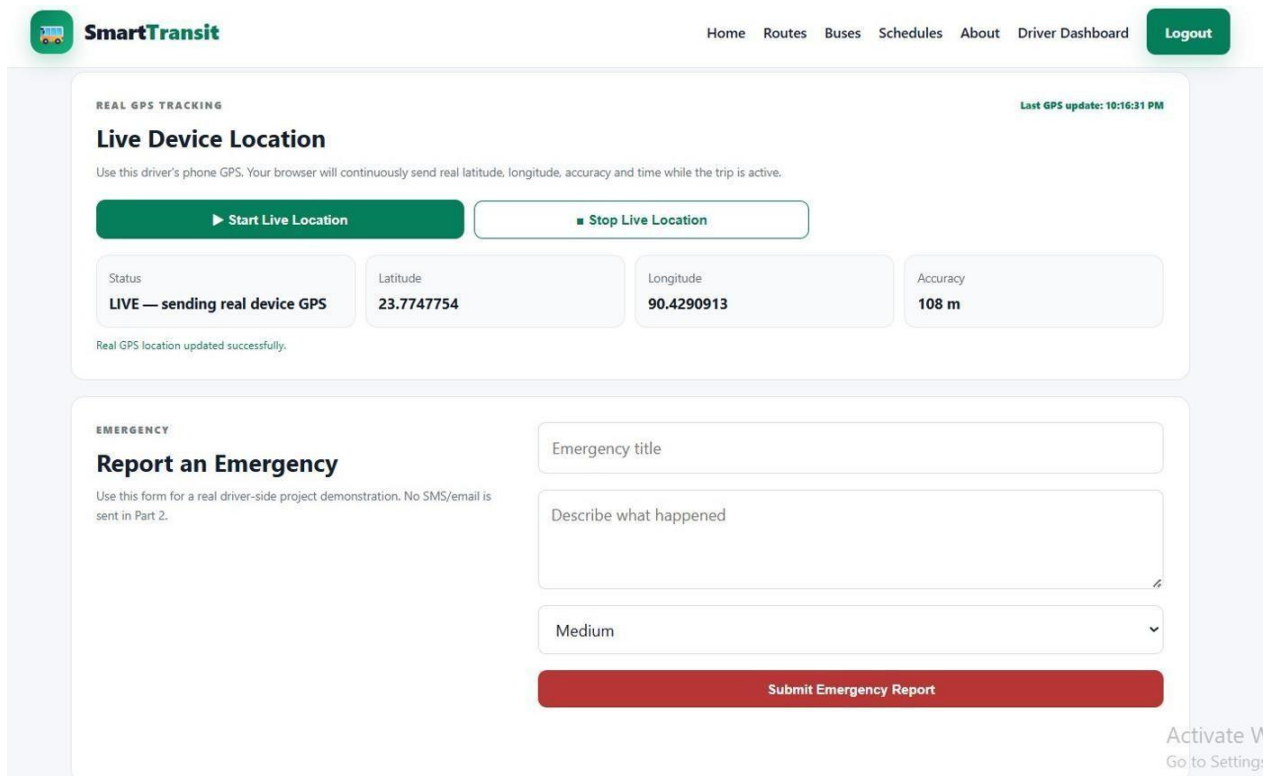
**Live Device Location**  
Use this driver's phone GPS. Your browser will continuously send real latitude, longitude, accuracy and time while the trip is active.

**Start Live Location** **Stop Live Location**

Last: 2026-08-19 00:29:25

Activate Wind  
Go to Settings to ac

Fig 9.8 - Driver login page



**SmartTransit** Home Routes Buses Schedules About Driver Dashboard Logout

REAL GPS TRACKING Last GPS update: 10:16:31 PM

### Live Device Location

Use this driver's phone GPS. Your browser will continuously send real latitude, longitude, accuracy and time while the trip is active.

▶ Start Live Location ■ Stop Live Location

|                                                 |                               |                                |                          |
|-------------------------------------------------|-------------------------------|--------------------------------|--------------------------|
| Status<br><b>LIVE — sending real device GPS</b> | Latitude<br><b>23.7747754</b> | Longitude<br><b>90.4290913</b> | Accuracy<br><b>108 m</b> |
|-------------------------------------------------|-------------------------------|--------------------------------|--------------------------|

Real GPS location updated successfully.

---

EMERGENCY

### Report an Emergency

Use this form for a real driver-side project demonstration. No SMS/email is sent in Part 2.

Emergency title

Describe what happened

Medium

**Submit Emergency Report**

Activate Windows  
Go to Settings to activate Windows.

Fig 9.9 - Driver share bus location

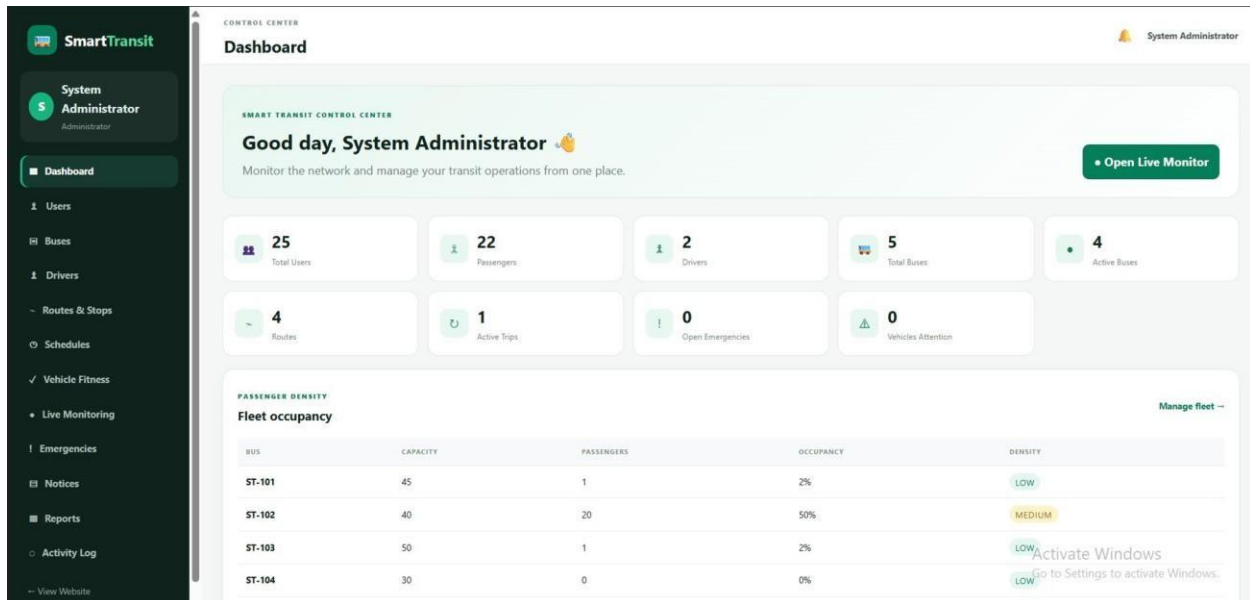


Fig 9.10 - Admin Dashboard

**SmartTransit**

**System Administrator**

**User Management**

Search, filter and control registered accounts.

Search name, email or phone All roles All statuses Filter

| USER                                   | PHONE     | ROLE      | STATUS | CREATED             | ACTIONS    |
|----------------------------------------|-----------|-----------|--------|---------------------|------------|
| Mohtasim Haque<br>mohtasim34@gmail.com | 017121947 | passenger | active | 2026-08-19 22:06:01 | Deactivate |
| Labiba akter<br>labiba0963@gmail.com   | 030838    | passenger | active | 2026-08-19 00:52:57 | Deactivate |
| Akter hassain<br>akter963@gmail.com    | 98967     | passenger | active | 2026-08-19 00:52:14 | Deactivate |
| Jahidul Islam<br>jahidul2@gmail.com    | 2983      | passenger | active | 2026-08-19 00:51:29 | Deactivate |
| Morlum Akter<br>morlum1234@gmail.com   | 39336     | passenger | active | 2026-08-19 00:50:30 | Deactivate |
| Siraj Mollah<br>siraj03756@gmail.com   | 09848     | passenger | active | 2026-08-19 00:49:46 | Deactivate |
| Hekkar alam<br>h3336@gmail.com         | 9373      | passenger | active | 2026-08-19 00:48:30 | Deactivate |

Activate Windows  
Go to Settings to activate Windows.

Fig 9.11 -User Management By Admin

**SmartTransit**

**System Administrator**

**Bus Management**

Maintain the existing fleet records and service status.

**Add Bus**

Bus Number: ST-101

Registration Number:

Bus Type: Standard

Capacity: 45

Service Status: active

☐ Women-only service

**Add Bus**

**Fleet list** 5 buses

| BUS                 | TYPE     | CAPACITY | PASSENGERS | OCCUPANCY | DENSITY | STATUS      | WOMEN-ONLY | ACTIONS     |
|---------------------|----------|----------|------------|-----------|---------|-------------|------------|-------------|
| ST-105<br>DHAKA-105 | Standard | 45       | 0          | 0%        | LOW     | active      | No         | Edit Toggle |
| ST-104<br>DHAKA-104 | Mini Bus | 30       | 0          | 0%        | LOW     | maintenance | No         | Edit Toggle |
| ST-103<br>DHAKA-103 | Standard | 50       | 1          | 2%        | LOW     | active      | Yes        | Edit Toggle |
| ST-102<br>DHAKA-102 | AC Bus   | 40       | 20         | 50%       | MEDIUM  | active      | No         | Edit Toggle |
| ST-101<br>DHAKA-101 | Standard | 45       | 1          | 2%        | LOW     | active      | No         | Edit Toggle |

Activate Windows  
Go to Settings to activate Windows.

Fig 9.12 -Bus Management By Admin

**SmartTransit**

System Administrator

Dashboard

Users

Buses

**Drivers**

Routes & Stops

Schedules

Vehicle Fitness

Live Monitoring

Emergencies

Notices

Reports

Activity Log

View Website

CONTROL CENTER

### Driver Management

Manage driver accounts and assign the existing bus/route relationships.

**STAFF**

#### Add Driver

Name

Email

Phone

Password

Status: active

Assigned Bus: No assignment

Assigned Route: No assignment

#### Drivers

| DRIVER                                        | STATUS | ASSIGNMENT    | ACTIONS     |
|-----------------------------------------------|--------|---------------|-------------|
| <b>Sabir Rafy</b><br>sabir1927@gmail.com      | active | Unassigned    | Edit Toggle |
| <b>Demo Driver</b><br>driver@smarttransit.com | active | ST-101 - R-01 | Edit Toggle |

Activate Windows  
Go to Settings to activate Windows.

Fig 9.13 - Driver Management By Admin

**SmartTransit**

System Administrator

Dashboard

Users

Buses

Drivers

Routes & Stops

Schedules

**Vehicle Fitness**

Live Monitoring

Emergencies

Notices

Reports

Activity Log

View Website

CONTROL CENTER

### Vehicle Fitness

Latest driver condition reports for every bus.

**FLEET HEALTH**

| BUS    | SERVICE STATUS | CONDITION    | LATEST REPORT                   | DRIVER      | MAINTENANCE |
|--------|----------------|--------------|---------------------------------|-------------|-------------|
| ST-101 | active         | Good         | 2026-08-19 00:28:55<br>(Recent) | Demo Driver | OK          |
| ST-102 | active         | Not reported | ---                             | ---         | OK          |
| ST-103 | active         | Not reported | ---                             | ---         | OK          |
| ST-104 | maintenance    | Not reported | ---                             | ---         | OK          |
| ST-105 | active         | Not reported | ---                             | ---         | OK          |

Activate Windows  
Go to Settings to activate Windows.

Fig 9.14 -Vehicle Fitness

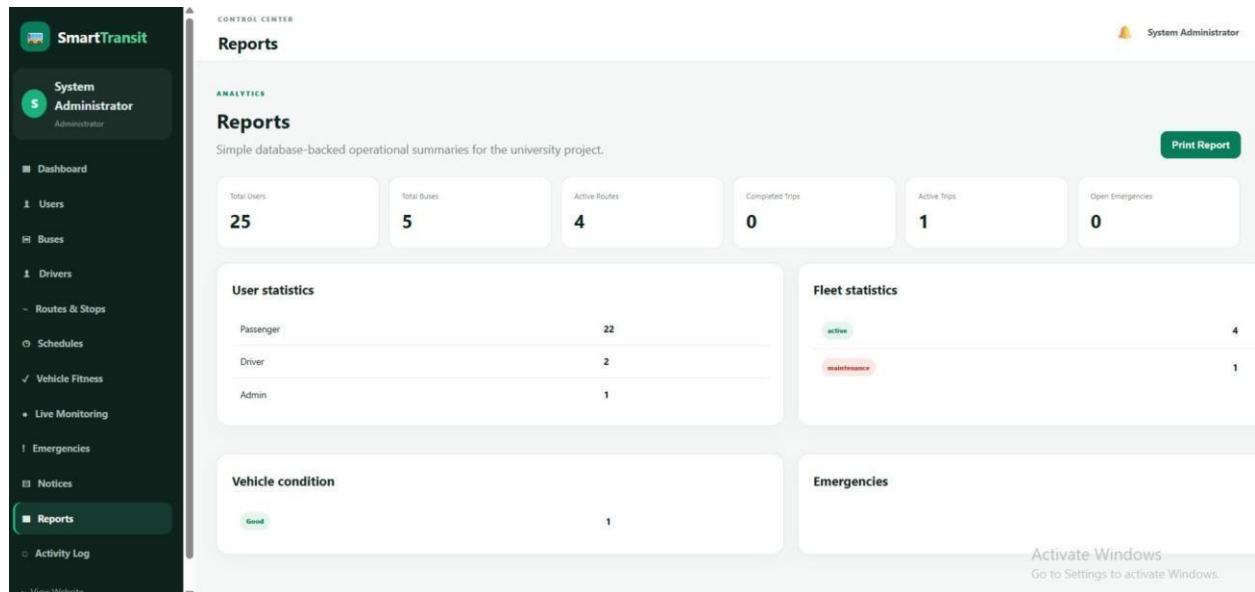


Fig 9.15 – Reports

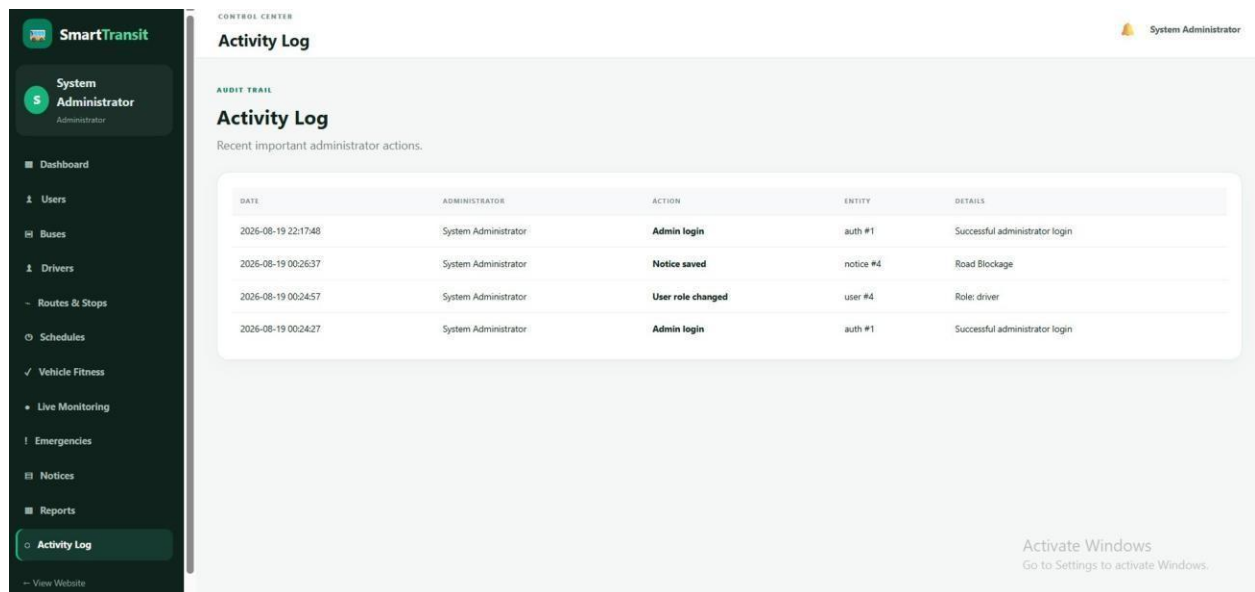


Fig16- Activity log

## 10. Limitation

1. Dependence on a stable internet connection for real-time GPS tracking.
2. GPS signals may become inaccurate in areas with poor network coverage.
3. The system depends on drivers regularly updating vehicle information.
4. Crowd density information may not always be completely accurate.
5. The current system focuses only on buses and does not include other public transportation services.
6. Integrating the system with existing transportation infrastructure may require additional resources and government support.
7. The system may face security and privacy challenges when handling user and location data

## 11. Future Improvements

1. Add AI-based traffic prediction to estimate delays more accurately.
2. Integrate online ticket booking and digital payment systems.
3. Expand the system to include trains, metro services, and other public transportation.
4. Add multilingual support to make the application more accessible.
5. Improve crowd analysis using IoT sensors and real-time passenger data.
6. Develop a dedicated mobile application for Android and iOS users.
7. Introduce a recommendation system that suggests the fastest and least crowded routes.

## 13. conclusion:

The Smart Transit Navigation System has designed to improve public transportation by providing a centralized platform for passengers, drivers and transport authorities. The system integrates real-time GPS tracking, route management, crowd monitoring, emergency reporting, vehicle fitness monitoring, and notification services into a single platform. The proposed system can improve communication between transportation stakeholders, reduce uncertainty for passengers, and help authorities monitor transportation services more efficiently. Although some limitations still exist, the system provides a practical foundation for developing a more modern, reliable, and user-friendly public transportation system in the future..